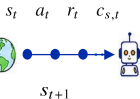


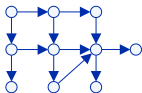
# 1 Rollout



Collect transitions  
( $s_t, a_t, r_t, c_{s,t}, s_{t+1}$ )

# 2 Flow inputs

Rollout graph



flow  $\hat{F}$

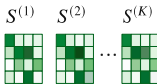


adv.  $A$



cost  $C_k$

# 3 Soft / hard



$S$



soft:  $b_k = B_k/L$   
hard:  $F_H = 0$

# 4 Flow projection

$$F^\star = \text{proj}_\epsilon(\hat{F}; b, H)$$

reroute  $\hat{F} \rightarrow F^\star$

$K_{ij} = e^{-D_{ij}/\epsilon}$  flow cost target  $F^\star$



target flow only

prices  $\lambda^\star$

# 5 Priced PPO

$$\lambda = \bar{\lambda}^\star + \beta$$

$$\lambda = \bar{\lambda}^\star + \beta$$

priced advantage

$$\tilde{A} = A_r - \lambda \cdot A_c$$

PPO / TRPO

$\pi_{t+1}$

